Aspiration Traps

Evidence from tertiary education in Colombia

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Inequality in Latin America

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  - Most of the literature on Latin America and Colombia's inequlity is based on this argument ((De Barros and Ferreira, 2009), (Gasparini et al. (2010)), Cruces et al. (2011)).
- However, this is only a part of the story, in fact **this approach fails to explain why some students drop out of school, or let go financial aid for post-secondary education**.

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  - Lack of experiences and interaction with successful role models may keep poor children from growing more ambitious aspirations.
  - Aspirations might be too far from their current living standard, which induces frustration.
- Both of these eventually lead to under-investment and relatively low adult outcomes.

Previous literature

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- The main empirical problem here is that aspirations are potentially endogenous.
- Macours and Vakis (2009) is so far the only paper that identifies the impact of aspirations on productive and human capital investment using a CCT randomized trial.

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  - Instrumental variables are used to identify the causal effect.
  - This non experimental approach allows to use a very large sample of Colombian students that graduated from tertiary education.

Educational aspirations and achievements

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- High school students are asked about the major they want to pursue, and the institutions of higher education they have in mind.
- Students apply to a specific major and it is usually hard to change afterwards given the rigidity of the curriculum.

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- Matched sample is not representative of high school students: regression results are only valid for students who graduated from tertiary education.

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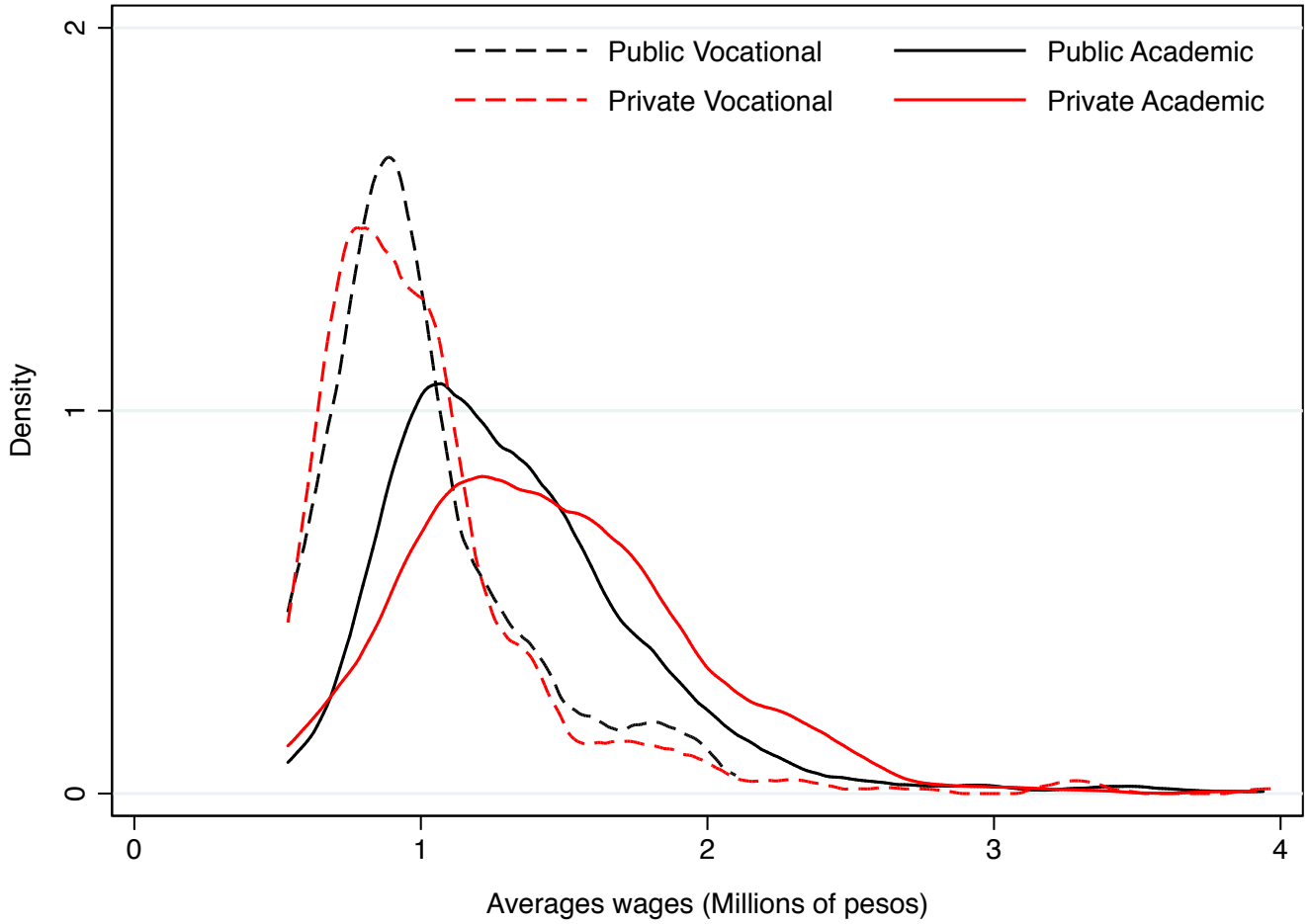
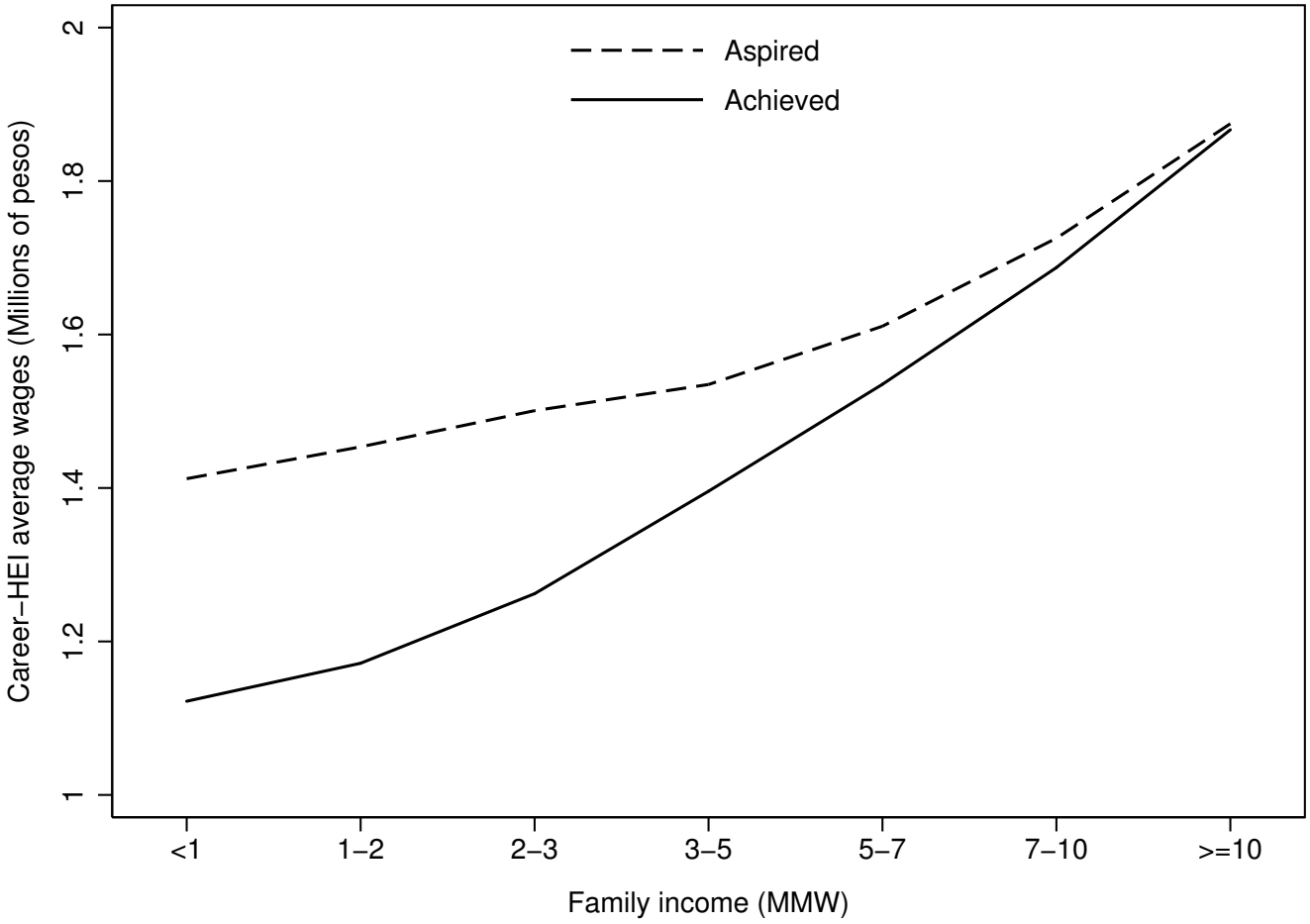
- Ministry of Education: College graduation records merged to social security datasets to compute the 2012 average wages of students that graduates in 2011.
- Merged at individual level to students aspiration and achieved career-college by:
  - Vocational-academic education
  - Field (56 categories)
  - Higher education institution (264)
  - municipality

Career-college average wage of recent graduates



Aspired and achieved career-college average wage by family income



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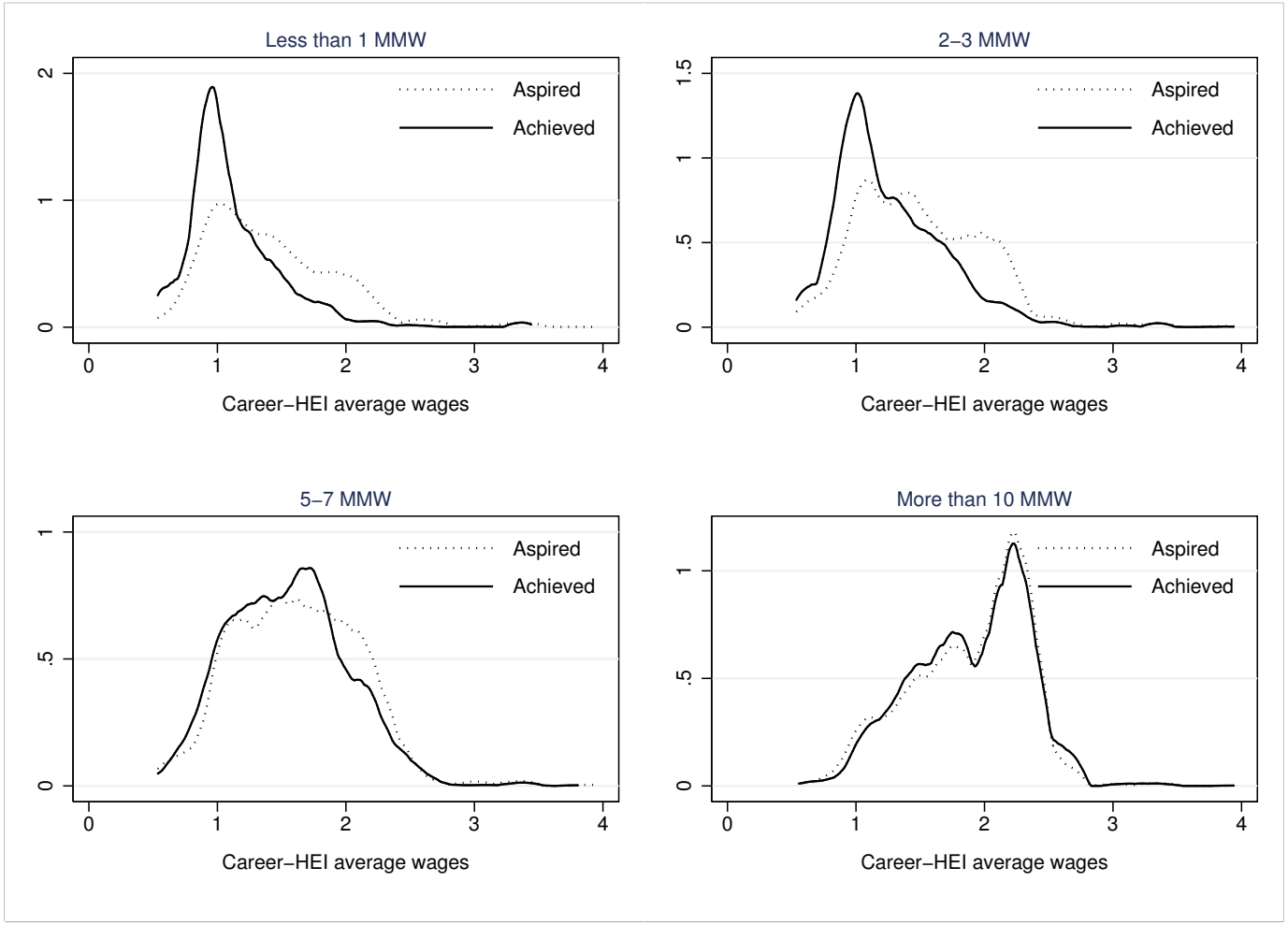
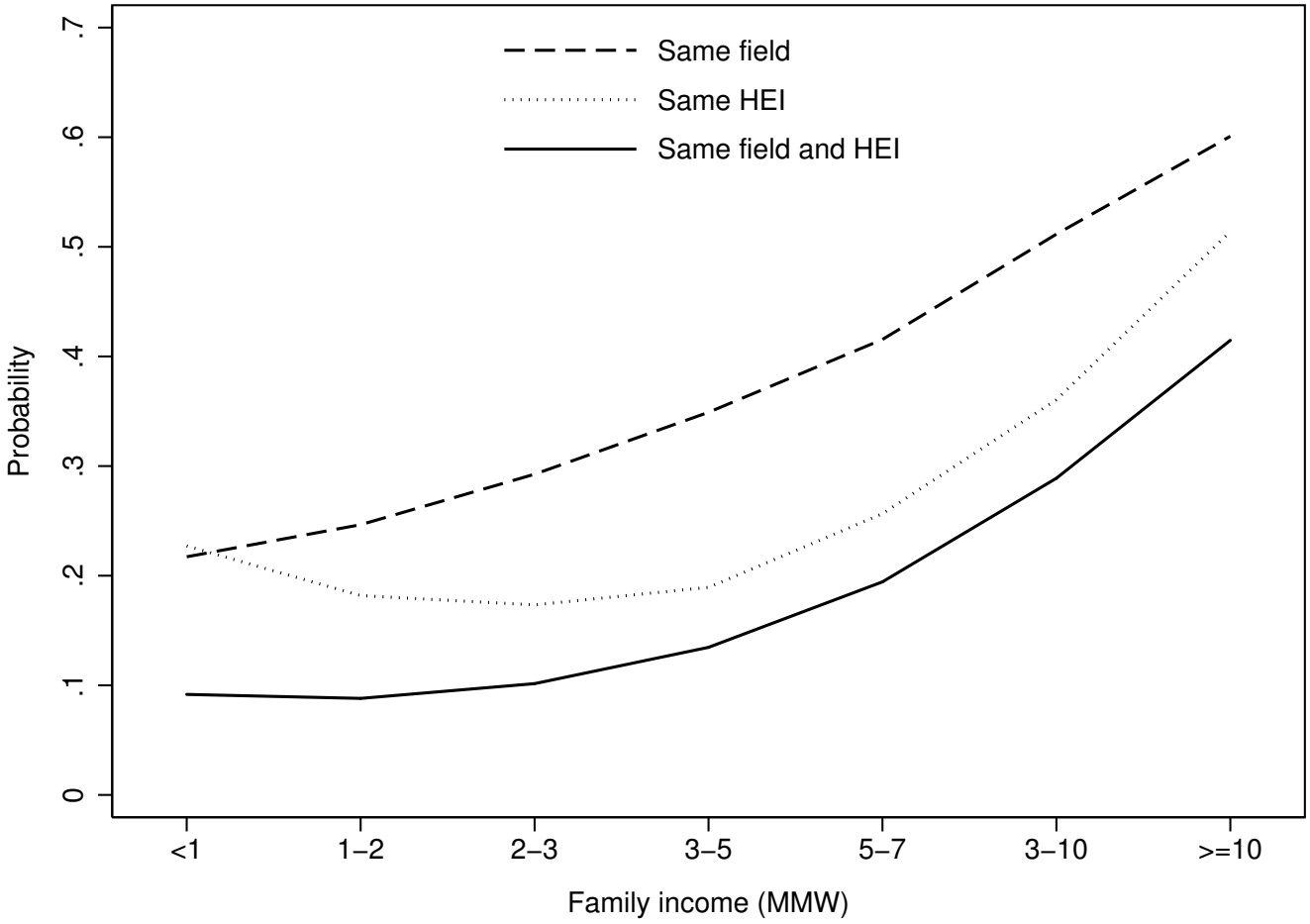
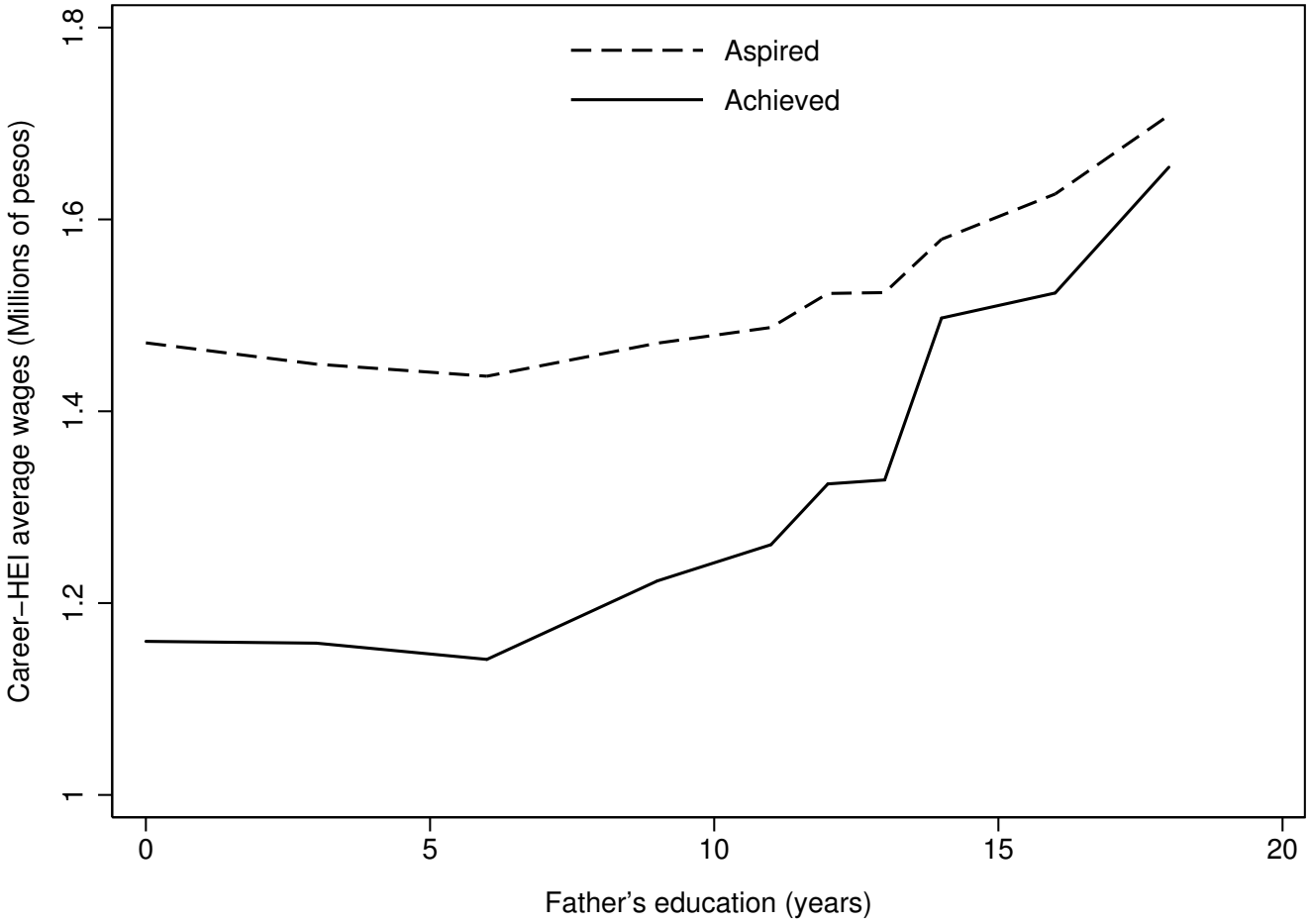


Probability of achieving aspired field and Higher education institution (HEI) by family income



Aspired and achieved career-college average wage by father's education



Identification strategy: IV

Model:

$$y_{ij} = \beta_1 Z_{ij} + \beta_2 X_{ij} + u_{ij} \tag{1}$$

$$Z_{ij} = \gamma_1 I_{ij} + \gamma_2 X_{ij} + v_{ij} \tag{2}$$

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Exogenous covariates (X_{ij}) include:

- Individual: gender, age, ethnicity.
- Family: household size, parents education (quadratic), family income.
- School: private, calendar, mixed, single shift, tuition fees.
- Municipality: population, urbanization, poverty and homicide rates.

Identification strategy

Model:

$$y_{ij} = \beta_1 Z_{ij} + \beta_2 X_{ij} + u_{ij} \tag{3}$$

$$Z_{ij} = \gamma_1 I_{ij} + \gamma_2 X_{ij} + v_{ij} \tag{4}$$

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Two specifications:

- $Z_{ij} = Asp_{ij}$: Only aspirations as endogenous
- $Z_{ij} = [Asp_{ij}, Tests_{ij}]$: Aspirations and test scores as endogenous

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Available instruments (I_{ij}):

- Peers' average aspiration career-HEI wages (quadratic)
- Peers' parents education (quadratic)
- Percentage of young adults studying in hometown (20 years)

Only aspirations as endogenous

	Achieved career-college average wage			First stage	
		IV		Aspired averageg wage	
	OLS	(1)	(2)	(1)	(2)
Average wage	0.169	0.062	0.056		
	(0.007)***	(0.015)***	(0.014)***		
Mathematics	0.046	0.044	0.046	0.010	0.010
	(0.002)***	(0.002)***	(0.002)***	(0.002)***	(0.002)***
Language	0.035	0.036	0.036	0.005	0.005
	(0.002)***	(0.002)***	(0.002)***	(0.002)*	(0.002)**
Peers' aspired average wage				12.000	13.440
				(1.087)***	(0.999)***
Peers' aspired average wage				(0.3880)	(0.4390)
(Squared)				(0.038)***	(0.035)***
Additional instruments				Yes	No
First stage F				1001.60	3721.7
p-value				(0.000)***	(0.000)***
Hansen J test		163.878	98.775		
p-value		(0.000)***	(0.000)***		
N	39,279	35,797	39,207	35,797	39,207

* Significant at 10% ; ** significant at 5% ; *** significant at 1%.
Notes: Coefficient standard errors in parenthesis. Estimations assume clustered errors at school level. Additional instruments include peers' parents education (quadratic) and Percentage of 20 years old that are studying in hometown.

Aspirations and test scores as endogenous

	Achieved career-college average wage			First stage	
	OLS	IV	Aspired averges wages	Mathematics	Language
Average wage	0.169	0.136			
	(0.007)***	(0.034)***			
Mathematics	0.046	0.586			
	(0.002)***	(0.167)***			
Language	0.035	-0.274			
	(0.002)***	-0.232			
Peers' aspired average wage				-24.660	-15.810
				(4.176)***	(2.932)***
Peers' aspired average wage (Squared)				0.863	0.556
				(0.147)***	(0.103)***
Additional instruments				Yes	Yes
First stage F			998.560	17.880	10.410
p-value			(0.000)***	(0.000)***	(0.000)***
Hansen J test		3.251			
p-value		(0.516)			
N	39,279	35,797	35,797	43,164	43,164

* Significant at 10% ; ** significant at 5% ; *** significant at 1%.
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  - To verifiy this, we use a Hausman endogeneity test for these two endogenous variables.
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- Results indicate that **higher aspirations and math test scores have a positive effects on educational achievements**.